
Standardization documents — Document metamodel

Documents de normalisation — Métamodèle des documents

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Contents

Page

Foreword	v
0 Introduction	vi
0.1 General	vi
0.2 Challenges	vi
0.3 Data-centric approach	vi
0.4 Compatibility	vii
0.5 Conventions	vii
1 Scope	1
2 Normative references	1
3 Terms and definitions	1
4 Architecture	3
4.1 General	3
4.2 Dependent data models	3
4.3 Structure	3
4.4 Underlying document model	3
4.5 Bibliographic and citation models	4
4.6 Hierarchical containment	4
4.7 Document model specialization	4
5 StandardDocument	5
5.1 StandardDocument	5
5.2 BoilerplateType	6
6 Document metadata elements	7
6.1 Purpose	7
6.2 Structure	7
6.3 BibData	7
6.4 BibDataExtensionType	7
6.5 StandardBibData	8
6.6 StandardBibDataExtensionType	8
6.7 StructuredIdentifierType	8
6.8 EditorialGroupType	10
6.9 TechnicalCommitteeType	10
6.10 IcsType	10
6.11 StandardBibDataExtensionType	10
7 Section elements	11
7.1 Purpose	11
7.2 Structure	11
7.3 Clause sections	12
7.4 Content sections	12
7.5 Annex sections	12
7.6 Boilerplate sections	12
7.7 Category label	13
7.8 Restrictions on hanging elements	13
7.9 StandardSection	13
7.10 StandardHierarchicalSection	13
7.11 ClauseSection	14
7.12 AnnexSection	14
7.13 StandardReferencesSection	14
7.14 ClauseHierarchicalSection	15
7.15 StandardContentSection	15
7.16 NormativeType	15

8	Citation and reference elements	16
8.1	Purpose	16
8.2	StandardReferencesSection	16
8.3	Reference item	16
8.4	Citation	16
9	Terms and definition elements	16
9.1	Purpose	16
9.2	General	17
9.3	DefinitionSection	17
9.4	DefinitionCollection	17
9.5	Definition	17
9.6	TermsSection	17
9.7	TermCollection	17
9.8	Term	18
9.9	RelatedTerm	19
9.10	RelatedTermType	19
9.11	TermSource	19
9.12	TermSourceStatus	20
9.13	Nested sections	20
10	Inline elements	20
10.1	ReferenceToTermbase	20
10.2	StandocReferenceToldElement	21
10.3	StandardUnorderedList	21
10.4	UnorderedCheckableListItem	21
10.5	StandardDefinitionList	22
11	Block	22
11.1	StandardNoteBlock	22
11.2	StandardTableBlock	23
11.3	TableCol	23
11.4	StandardFigureBlock	24
11.5	StandardSourcecodeBlock	24
11.6	StandardFormulaBlock	24
11.7	StandardExampleBlock	25
11.8	StandardParagraphBlock	25
11.9	StandardQuoteBlock	25
11.10	StandardAdmonitionBlock	25
11.11	AmendBlock	26
11.12	AutoNumber	26
11.13	ImageMapBlock	27
11.14	ChangeType	27
11.15	ElementName	27
11.16	SvgMapBlock	28
11.17	SvgTargetType	28
11.18	ImageMapAreaType	28
11.19	ImageMapAreaTypeType	29
11.20	ImageMapCoords	29
11.21	ImageMapRadius	29
12	Data types	30
Annex A (informative)	Profile: ISO standardization deliverables	31
Annex B (informative)	Profile: CalConnect standardization deliverables	42
Annex C (informative)	Concept and concept collection models	46
Bibliography		54

Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standardization bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the Introduction and/or on the ISO list of patent declarations received (see www.iso.org/patents).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation on the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT) see the following URL: www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 154, *Processes, data elements and documents in commerce, industry and administration*.

This is the first edition of this document.

0 Introduction

0.1 General

Standardization documents are documents created to express a set of normalized requirements, often themselves created through a process of standardization.

Standardization documents play a crucial role in today's activities. Rule-based enforcements from the organization of markets, conformance of products to the compliance of regulation and requirements, all depend on content expressed through standardization documents.

0.2 Challenges

The process of generating standardization documents is increasingly problematic due to the increased work burden on standardization bodies.

Often, concerns are raised over both the efficiency and the effectiveness of the workflows around generating standardization documents in those bodies and elsewhere. Some practices include:

- Rigorous and detailed rules around document structure articulated by standardization bodies are being manually validated by the editorial teams, with mixed and often inconsistent results.
- Editing of documents substantially revolves around standardizing use of particular software, instead of requirements, with all the problems that conflating document structure and presentation introduce.
- The authoring and editorial processes spend significant amounts of time on formatting rather than document content, and is vulnerable to error.
- Existing exchange mechanisms for documents have been claimed to focus on document appearance, rather than document structure, and so do not alleviate the formatting burden on the editorial process.
- Standardization bodies often demand vastly different workflows that take different inputs. Submitting identical content or cross-publishing to more than one standardization body requires replicating a significant amount of work.

0.3 Data-centric approach

The *StandardDocument* document metamodel aims to treat a standardization document as a set of standardized data structures, used to enable an end-to-end standardization document production process. This process is designed to focus on semantics and encompasses core similarities between standardization documents of different standardization bodies.

This approach has the following benefits:

- The approach enables an end-to-end, author-to-publishing workflow, with well-managed data artefacts along each stage, that editorial bodies can exercise appropriate control over. This allows document management processes to be centered around data.
- The semantic nature of the document metamodel (and its specialized models) abstracts away presentation, such as layout and formatting. This allows the authoring and editorial to focus on content.
- The structural and content requirements around standardization documents can be fully captured in the document model, and can therefore be subject to formal validation.
- The document model can be highly similar across different standardization documents, and therefore the content in such documents can be easily repurposed for different standardization bodies with minimal effort.

0.4 Compatibility

The document metamodel in this standard has been designed to reflect the needs and expressiveness of existing standardization processes, requirements and outputs of the following standardization bodies:

- ISO and IEC (ISO/IEC DIR 1, ISO/IEC DIR 2 ISO)
- ITU (TTP, ATP)
- CalConnect (csd CC/DIR 10001:2019, csd CC/DIR 10002:2019)
- IETF (IETF RFC 7749, IETF RFC 7991)
- OGC (OGC 05-020r26)
- NIST Special Publications
- BIPM
- IHO

In many cases, the document model also enables alignment with capabilities of existing document production tools. The document metamodel has already been adopted by a number of standardization bodies listed above.

0.5 Conventions

In this document, italics is used to name document models, such as *StandardDocument*. Monospace is used to refer to elements of document models, such as `InlineElement`.

Standardization documents — Document metamodel

1 Scope

This document provides a document metamodel for standardization documents and describes its alignment to standardization processes.

The following aspects of the document metamodel are out of scope:

- implementation of the document metamodel into a document model;
- derivation and mapping of the reference model to representation formats (such as for serialization); and
- prescription of profiles of the document metamodel for particular standardization bodies.

Example profiles provided for ISO and CalConnect standards in [Annex A](#) and [Annex B](#) are presented for informative reasons.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 639 (all parts), *Code for the representation of names of languages*

ISO 8601-1, *Date and time — Representations for information interchange — Part 1: Basic rules*

ISO/IEC 10118 (all parts), *Information technology – Security techniques*

ISO/IEC 14888 (all parts), *Information technology – Security techniques*

ISO 15924 (all parts), *Information and documentation – Codes for the representation of names of scripts*

ISO/IEC Guide 2 (all parts), *Standardization and related activities – General vocabulary*

RFC 3986, *Uniform Resource Identifier (URI): Generic Syntax*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <http://www.iso.org/obp>
- IEC Electropedia: available at <http://www.electropedia.org>

3.1

standardizing body

body that has recognized activities in standardization

[SOURCE: ISO/IEC Guide 2, 4.3]

3.2

document

medium with information recorded on or in it

[SOURCE: ISO/IEC Guide 2, 3.1 NOTE 2]

3.3

standardization document

document that provides rules, guidelines or characteristics for activities or their results

Note 1 to entry: Documents such as standards, technical specifications, codes of practice and regulations are considered part of this term.

Note 2 to entry: This term closely corresponds to “normative document” defined in ISO/IEC Guide 2, 3.1.

3.4

class

structure containing a description of an entity in terms of its components

3.5

subclass

class ([3.4](#)) which inherits from another class its component descriptions, and optionally adds to them its own component descriptions

3.6

document metamodel

formal specification of the structure of a document model in terms of its components and their arrangement, expressed through *classes* ([3.4](#))

3.7

document model

formal specification of the structure of a document in terms of its components and their arrangement, expressed through *classes* ([3.4](#))

3.8

standards model

document model ([3.7](#)) for *standardization documents* ([3.3](#))

3.9

paragraph

subdivision of running text, normally run on throughout, that is separated from text before and after by a change of line and stands below any chapters or *sections* ([3.12](#))

[SOURCE: ISO 5127:2017, 3.5.8.07]

3.10

block

grouping of text at the *paragraph* ([3.9](#)) level

3.11

inline element

grouping of text that can be contained within a *paragraph* ([3.9](#)), including plain strings

3.12

section

hierarchical subdivision of a document, consisting of one or more *blocks* ([3.10](#)), and/or one or more sections

3.13

identifier

a character, or group of characters, used to identify or name an item of data and possibly to indicate certain properties of that item

[SOURCE: IEC 60050-714:1992, 714-21-07]

3.14 term

expression with a particular meaning that is part of the specialized vocabulary of a field

4 Architecture

4.1 General

The *StandardDocument* document metamodel is a semantically oriented data model that depends on an underlying document model for implementation.

StandardDocument is closely aligned with the *BasicDocument* model (CC 36010), but it can be implemented on top of any document model that is compatible with the *BasicDocument* model or satisfies the requirements described in this document.

4.2 Dependent data models

The implementation of the document metamodel relies on the availability of the following data models:

- A document data model that supports modeling of textual content at the block and inline levels, as well as modifications to the document itself.
- Data models for representing bibliographic information and citations for a document.

In this document, the *BasicDocument* document model from CC 36010 is used as an exemplary document model, the *BibliographicItem* and its related data models from CC 6900 are used for representing bibliographic and citation models.

4.3 Structure

The *StandardDocument* document metamodel is expressed hierarchically in the following structure:

- Document representation: the underlying document model, described as *BasicDocument* in this document, is used to express the structure of generic documents (with specific requirements detailed in the *BasicDocument* model from CC 36010);
- Standardization document structure: the *StandardDocument* model is specialized from *BasicDocument*, conveying the structure particular to standardization documents as a class.
- Specific types of standardization documents: the *StandardDocument* model is further specialized into models specific to particular standards. Specific standards models can form the basis in turn of other standards models.

4.4 Underlying document model

The underlying document model must provide the following classes of document element data models:

- Sections. A grouping of elements that can be shown under the table of contents (see the *Section* models in CC 36010)
- Blocks. A grouping of paragraphs or paragraph-levels of text (see *Block* models in CC 36010)
- Inline elements. Groupings of text smaller than a paragraph, including plain strings. (see *InlineElement* and related models in CC 36010)

4.5 Bibliographic and citation models

Standardization documents often contain references to and citations of external information resources, as well as bibliography sections.

The *StandardDocument* model depends on data models that can fully represent:

- metadata of referenced information resources;
- presentation of a referenced information resource; and
- locality of a citation.

For example, if a standardization document needs to adhere to the citation guidelines of ISO 690, data models that facilitate these rules such as CC 6900 should be used.

The *BibliographicItem*, *Citation* and related models from CC 6900 are used in the reference metamodel for its expression of bibliographic references and citations.

4.6 Hierarchical containment

In the document metamodel, the classes are in a strict hierarchical relation:

- Documents consist of sections, which consist of blocks, which consist of inline elements.
- Sections can be nested within sections (e.g. clauses and subclauses).
- Blocks can be nested within blocks (e.g. nested lists).
- Inline elements can be embedded within other inline elements (e.g. bold with italics).

Sections in the *StandardDocument* model are not expected to be siblings of blocks, nor blocks of inline elements.

EXAMPLE A list (block) is not expected to occur next to inline text within a paragraph. For that reason, paragraphs cannot contain other block elements, such as lists or tables.

NOTE This constraint is adopted from CC 36010, and is not present in XML-based schemas like HTML, TEI-C, or DocBook. It confers the advantage of having a significantly streamlined document model. While this limits some expressive potential, the difference is minor and semantically desired, particularly with regards to the unambiguous rendering of paragraphs.

4.7 Document model specialization

StandardDocument can be specialized in the following ways:

- specializing classes from the *BasicDocument* models;
- specializing classes from the *StandardDocument* models;
- adding new classes;
- changing attributes of a class.

The specialization method is not restricted to adding attributes, as is the case in typical entity subclassing; it can also include removing attributes from a class, changing their obligation and cardinality, and changing their type, including changing enumerations. Attributes can be overruled at any level; for example, standards-specific models routinely enhance the bibliographic model at the base of the hierarchy.

For reasons of clarity, however, renaming classes and attributes should be avoided in specialization.

EXAMPLE The document model for the Chinese National Standard (“GB Standard”), as described in Chinese Standard GB/T 1.1-2009, is structurally modelled after ISO documents, explicitly described as a localization of ISO/

IEC Directives Part 2, 2004. Therefore, the Chinese National Standard model is best expressed as a specialization of the ISO Standards model.

5 StandardDocument

5.1 StandardDocument

The *StandardDocument* model is used to represent a standardization document.

Typically, a standardization document contains at least the following data elements:

- Metadata information;
- Clauses and subclauses;
- Bibliographies;
- Annexes if applicable.

The exact component requirements of a standardization document can differ widely from one standardization body to the next. Specialization is necessary to adopt the *StandardDocument* model for such standardization bodies.

The *StandardDocument* model extends the *BasicDocument* modelling of the document by requiring specific types of sections.

Table 1 — StandardDocument attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
bibdata	Bibliographic description of the document itself, expressed in the Relaton model.	M	1	StandardBibData
misc	Extension point for extraneous elements that need to be added to standards document from other schemas, e.g. UnitsML.	O	1	MiscContainer
boilerplate	An optional <i>boilerplate</i> section, intended to appear at the front of the document. It consists of content addressing <i>copyright</i> , <i>license</i> , <i>legal</i> , and <i>feedback</i> concerns.	O	1	BoilerplateType
preface	Zero or more optional <i>preface</i> sections.	O	N	StandardHierarchicalSection
sections	One or more <i>sections</i> .	M	N	StandardHierarchicalSection
annex	Zero or more <i>annexes</i> .	O	N	StandardHierarchicalSection

Table 1 — StandardDocument attributes

(continued)

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
bibliography	Zero or more <i>bibliographies</i> .	0	N	StandardReferencesSection
indexsect	Index of a standards document.	0	N	StandardHierarchicalSection
type	Type of standards document representation given in this class instance. Set to “standard” for all standards documents.	M	1	StandardDocumentType
version	Version number of the schema used for this standards document.	M	1	String

5.2 BoilerplateType

Content addressing legal and licensing concerns around the document, outside of the main flow of document content.

Table 2 — BoilerplateType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
copyright	Preset templated text provided by the standardization body, describing the copyright status of the document.	0	1	HierarchicalSection
license	Preset templated text provided by the standardization body, providing the licensing terms for the document content.	0	1	HierarchicalSection
legal	Preset templated text provided by the standardization body, providing the legal constraints and considerations around use of the document.	0	1	HierarchicalSection
feedback	Preset templated text provided by the standardization body, providing information on where feedback on the document may be addressed to.	0	1	HierarchicalSection

6 Document metadata elements

6.1 Purpose

Document metadata of a standardization document is used to provide basic information of the standardization document.

This includes data elements such as:

- title(s);
- document identifier(s);
- authorship and contribution information;
- publisher and distributor details;
- rights information;
- document development stage;
- relevant dates; and
- edition.

6.2 Structure

Document metadata in *StandardDocument* is handled using the `StandardBibliographicItem` class extended from the `BibliographicItem` class defined in CC 6900.

In standardization documents, certain data elements, such as document identifiers and document development stage information, are often highly specific to a standardization body.

Therefore, based on `BibliographicItem`, `StandardBibliographicItem` implements the following extension points:

- the document type (already provided in the *BasicDocument* model), it is assumed that particular specializations of the document metamodel will substitute their own enumerations of particular subclasses of document, which will be more granular;
- zero or more editorial committees responsible for producing the document (e.g. technical committees or subcommittees);
- zero or more classifications of the document following the International Classification for Standards (ICS) code (ISO International Classification for Standards);
- zero or more structured identifiers.

Other extension points may be added specific to the standardization body.

6.3 BibData

Bibliographic description of document, expressed through CC 6900.

6.4 BibDataExtensionType

Extension point for `Bibdata`, containing elaborations of description of current document specific to the publishing organization.

6.5 StandardBibData

The bibliographic description of a *StandardDocument*.

Table 3 — StandardBibData attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
ext	The extension point of the bibliographic description of a <i>StandardDocument</i> .	M	1	StandardBibDataExtensionType

6.6 StandardBibDataExtensionType

The extension point of the bibliographic description of a *StandardDocument*.

Table 4 — StandardBibDataExtensionType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
doctype	Classification of the <i>StandardDocument</i> that is treated as a distinct series by the standards defining organization, and that is rendered in a distinct manner.	M	1	String
doctype_abbreviation	Standard abbreviation for the doctype value used by the standards defining organization.	0	1	String
editorialgroup	Groups associated with the production of the standards document, typically within a standards definition organization.	0	1	EditorialGroupType
ics	Classifications of the document contents taken from the International Classification of Standards.	0	N	IcsType
structuredidentifier	Representation of the identifier for the <i>StandardDocument</i> , giving its individual semantic components.	0	N	StructuredIdentifierType

6.7 StructuredIdentifierType

Representation of the identifier for a *StandardDocument*, giving its individual semantic components.

Table 5 — StructuredIdentifierType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	Representation in the identifier of the type of standard document.	0	1	String
agency	Representation in the identifier of the agency responsible for the standard document.	M	N	String
class	Representation in the identifier of the class of standard document (as a subclass of the document type).	0	1	String
docnumber	Representation in the identifier of the (typically numeric) component uniquely identifying the document or standard. If a document includes parts or supplements, the docnumber identifies the document as whole, and not those document components.	M	1	String
partnumber	Representation in the identifier of the document part, if this is a document part.	0	1	String
edition	Representation in the identifier of the document edition, if this is a published document.	0	1	String
version	Representation in the identifier of the document version, which can include document drafts.	0	1	String
supplementtype	Representation in the identifier of the type of document supplement, if this is a document supplement.	0	1	String
supplementnumber	Representation in the identifier of the document supplement, if this is a document supplement.	0	1	String
year	Representation in the identifier of the year of publication or issuance of the document.	0	1	String
language	Representation in the identifier of the language of the document.	0	1	String

6.8 EditorialGroupType

A group associated with the production of the standards document, typically within a standards definition organization.

Table 6 — EditorialGroupType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
technicalcommittee	A technical committee associated with the production of the standards document.	M	N	TechnicalCommitteeType

6.9 TechnicalCommitteeType

Technical committee associated with the production of a standards document.

Table 7 — TechnicalCommitteeType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
number	Identifier of the technical committee (typically numeric).	O	1	Integer
type	Type of the technical committee, used in identifying the technical committee.	O	1	String
text	Name of the technical committee.	M	1	String

6.10 IcsType

Classification taken from the International Classification of Standards. ICS is defined by ISO here — <https://www.iso.org/publication/PUB100033.html>

Table 8 — IcsType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
code	Classification code taken from the ICS.	M	1	String
text	Text string associated with the classification code.	O	1	String

6.11 StandardBibDataExtensionType

The extension point of the bibliographic description of a *StandardDocument*.

Table 9 — StandardBibDataExtensionType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
doctype	Classification of the <i>StandardDocument</i> that is treated as a distinct series by the standards defining organization, and that is rendered in a distinct manner.	M	1	String
doctype_abbreviation	Standard abbreviation for the doctype value used by the standards defining organization.	0	1	String
editorialgroup	Groups associated with the production of the standards document, typically within a standards definition organization.	0	1	EditorialGroupType
ics	Classifications of the document contents taken from the International Classification of Standards.	0	N	IcsType
structuredidentifier	Representation of the identifier for the <i>StandardDocument</i> , giving its individual semantic components.	0	N	StructuredIdentifierType

7 Section elements

7.1 Purpose

A section represents a clause in a standardization document. Specifically, a section represents a logically coherent group of paragraphs and blocks that is meant to be shown in the table of contents.

7.2 Structure

The *StandardDocument* model specializes the *BasicSection* class of the *BasicDocument* model by as follows:

- All sections in a *StandardDocument* model are instances of the *StandardSection* class, which is a subclass of *BasicSection*.
- The *StandardSection* class indicates its status as normative or informative explicitly, through the *status* attribute. The normative status of individual sections in a standardization document is pertinent.

EXAMPLE The necessity of indicating normative or informative status is specified in ISO/IEC DIR 2 ISO for ISO deliverables.

There are two major types of sections, namely, clause sections and content sections.

7.3 Clause sections

Clause sections (`ClauseSection`) are numbered, and form part of the main flow of the document.

Clause sections are by default hierarchical (as an exception, see [Clause 9](#)). The subclauses of Clauses are themselves Clause Sections; they continue the hierarchical numbering of Clause Sections at deeper hierarchical levels.

7.4 Content sections

Content sections (`StandardHierarchicalSection`) are optionally not numbered, and lie outside the main flow of the document.

In *StandardDocument*, there are four distinctly named content sections. These content sections are designated sections as they recur in standardization documents and have distinct semantics:

- Abstract,
- Foreword,
- Introduction, and
- Acknowledgments.

Specializations of the model may define other such sections.

The subclauses of Content sections are `StandardSections`, and need not be numbered.

7.5 Annex sections

Annexes are implemented as `ClauseSections` in the *StandardDocument* metamodel.

They are, however, expected to be numbered differently, and to be rendered as separate sequences from clause sections.

7.6 Boilerplate sections

Boilerplate sections include recurring text that is included in all documents of its class by the standardization body, and that reflects various legal or policy requirements imposed on the document.

Boilerplate sections are implemented as `ContentSections` in the *StandardDocument* metamodel.

They are, however, expected to be placed and rendered separately from the Content sections of the document.

In *StandardDocument*, there are four distinctly named boilerplate sections. These content sections are designated sections as they recur in standardization documents and have distinct semantics:

- Copyright Statement
- License Statement
- Legal Statement
- Feedback Statement

Specializations of the model may define other such sections.

7.7 Category label

Both section types can have a category label, which allows them to be semantically classified into recurring categories, which may be processed and rendered differently. Indicative values include:

- executive-summary,
- errata,
- methods,
- requirements,
- tests,
- reagents, and
- tests.

7.8 Restrictions on hanging elements

Generally, in standardization documents there is a strong need to reference text in clauses unambiguously by clause number, which makes hanging paragraphs undesirable.

Therefore, the *StandardDocument* model requires “hanging paragraphs” to be avoided. In particular, its sections should contain either blocks or subsections, but not both.

NOTE The avoidance of “hanging paragraphs” is also a requirement in ISO/IEC DIR 2 ISO.

While this behavior is less pressing for Content sections (which are unnumbered), and Annexes (as exemplified in ISO/IEC DIR 2 ISO), it is nonetheless not recommended.

7.9 StandardSection

Section of a *StandardDocument*.

Table 10 — StandardSection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
status	Force of this clause in the standards document.	M	1	NormativeType
amend	Specification of machine-readable change outlined in this section, in a document amendment.	O	1	AmendBlock
number	Value of number to be used for numbering of section, overriding any autonumbering in rendering.	O	1	String

7.10 StandardHierarchicalSection

A hierarchical section containing other sections, within a *StandardDocument*.

Table 11 — StandardHierarchicalSection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
subsections	Subclauses of the section.	0	N	StandardSection

7.11 ClauseSection

A numbered clause constituting part of the main body of a *StandardDocument*.

Table 12 — ClauseSection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	Semantic class of the numbered clause.	0	1	String

7.12 AnnexSection

Section of document constituting an annex or appendix.

Table 13 — AnnexSection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
subsections	Subsections of the annex. Added explicitly because AnnexSection permits hanging paragraphs (top-level blocks are included in the section alongside subsections).	0	N	ClauseSection

7.13 StandardReferencesSection

The *StandardReferencesSection* class within *StandardDocument* is used to represent a bibliography section. It is used to collate references within the document, where there could be one or more of such sections within a document.

For example, some standardization documents differentiate normative or informative references, some split references into sections organized by concept relevance.

Similar to the *ReferencesSection* of the *BasicDocument* model, they are leaf nodes which contain zero or more bibliographical references (as modelled in Relaton), along with any prefatory text.

Table 14 — StandardReferencesSection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
normative	Whether the references in the current section are normative or	0	1	Boolean

Table 14 — StandardReferencesSection attributes

(continued)

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
	informative for the document.			
references	Bibliographic items cited in the document.	0	N	BibliographicItem
note	Annotations accompanying bibliographic items.	0	N	ParagraphBlock

7.14 ClauseHierarchicalSection

Clauses in which all textual content belongs in a strict clause hierarchy, such that no blocks of text are siblings to subclauses.

Table 15 — ClauseHierarchicalSection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
subsections	Subclauses which are mutually exclusive with text content.	0	N	ClauseHierarchicalSection

7.15 StandardContentSection

An unnumbered clause, outside the main body of text of a *StandardDocument*.

Table 16 — StandardContentSection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	The type of content provided in the section, used as a semantic classification.	0	1	String

7.16 NormativeType

Table 17 — NormativeType values

Name	Definition
unstructured	The force of a clause of text in a <i>StandardDocument</i> .
normative	The content in this section has normative effect.
informative	The content in this section has informative effect.

8 Citation and reference elements

8.1 Purpose

Standardization documents often cite other documents within, and as a result they usually contain one or more bibliography sections indicating its data sources. Bibliography sections may be differentiated as being normative or informative.

8.2 StandardReferencesSection

The `StandardReferencesSection` class within *StandardDocument* is used to represent a bibliography section. It is used to collate references within the document, where there could be one or more of such sections within a document.

For example, some standardization documents differentiate normative or informative references, some split references into sections organized by concept relevance.

Similar to the `ReferencesSection` of the *BasicDocument* model, they are leaf nodes which contain zero or more bibliographical references (as modelled in Relaton), along with any prefatory text.

Table 18 — StandardReferencesSection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
normative	Whether the references in the current section are normative or informative for the document.	0	1	Boolean
references	Bibliographic items cited in the document.	0	N	BibliographicItem
note	Annotations accompanying bibliographic items.	0	N	ParagraphBlock

8.3 Reference item

Each reference item is represented using the `BibliographicItem` class defined in CC 6900.

It is not necessary to specialize reference items amongst documents from different standardization bodies as a document can cite documents of any type.

8.4 Citation

Each citation is represented by the `Citation` class defined in CC 6900.

It is not necessary to specialize citations since it is simply a document data element reference linked to a reference item.

9 Terms and definition elements

9.1 Purpose

Standardization documents are used to convey normative information; to achieve so it is imperative to provide a standardized set of terms and definitions.

9.2 General

The *StandardDocument* model provides two further subclasses of clause sections (*ClauseSection*):

- Definition section, and
- Term section.

Both types of section can include prefatory text.

9.3 DefinitionSection

Definition sections consist of one or more definition lists (*DefinitionCollection*), are used to define symbols and abbreviations used in the remainder of the document.

They can also be used as glossaries, with simple definitions, in contrast to the more elaborate definitions given in terms sections.

In addition to allowing prefatory text per section, each definition list within each definition section can also be preceded by prefatory text.

Table 19 — DefinitionSection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
definitions	Definition list.	M	N	DefinitionCollection

9.4 DefinitionCollection

Collection of definitions, constituting the core content of a DefinitionSection.

Table 20 — DefinitionCollection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
definitions	Individual definition.	O	N	Definition

9.5 Definition

9.6 TermsSection

Term sections (*TermsSection*) give elaborated definitions of terms used in a standardization document.

NOTE The *TermsSection* definition fully aligns with the structure and requirements of the “Terms and definition” section given in ISO/IEC DIR 2.

Table 21 — TermsSection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
terms	Terms defined in the term section.	M	1	TermCollection

9.7 TermCollection

A collection of terms, constituting the main body of a Terms Section.

Table 22 — TermCollection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
terms	Individual terms.	0	N	Term

9.8 Term

A term (`Term`) represents a terminology entry with its definition.

NOTE The `Term` definition fully aligns with the structure and requirements of terms in ISO/IEC DIR 2, 16.6.

Table 23 — Term attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
id	An optional identifier for the term, to be used in cross-references.	0	1	String
preferred	One or more names under which the term being defined is canonically known.	M	N	LocalizedString
admitted	Zero or more names which are acceptable synonyms for the term being defined.	0	N	LocalizedString
related	Zero or more names which are related to the term being defined. Each has a type, indicating how the term is related; permitted values are compare (for “see also” references to terms), contrast (for terms that illuminate the term definition as what it is not), see (if this is a deprecated term, to reference the preferred term)	0	N	RelatedTerm
deprecates	Zero or more names which are deprecated synonyms for the term being defined.	0	N	LocalizedString
domain	An optional semantic domain for the term being defined, in case the term is ambiguous between several semantic domains.	0	1	LocalizedString
grammar	Zero or more abbreviations giving grammatical information about the term (e.g. part of speech, gender).	0	N	LocalizedString
definition	The definition of the term applied in the current document.	M	N	Definition

Table 23 — Term attributes

(continued)

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
note	Zero or more notes about the term.	0	N	ParagraphBlock
example	Zero or more examples of how the term is to be used.	0	N	ParagraphBlock
source	Zero or more bibliographical sources for the term. These include the origin of the term, which is its bibliographical citation (as defined in Relaton); the status of the definition (whether identical to the definition given in the origin cited, or modified); and, if the definition is modified, a description of the modification to the definition applied for this document.	0	N	TermSource

9.9 RelatedTerm

Term related to the current term.

Table 24 — RelatedTerm attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	Type of relation of term applicable.	M	1	RelatedTermType
content	Label of term related to the current term.	M	1	LocalizedString

9.10 RelatedTermType

Table 25 — RelatedTermType values

Name	Definition
unstructured	Type of relation between terms.
compare	The related term is mentioned as something the current term is comparable to.
contrast	The related term is mentioned as something the current term is to be contrasted with.
see	The related term is mentioned as something to be consulted as providing further information on the current term.

9.11 TermSource

The bibliographic source where a term is defined in the sense applicable in this *StandardDocument*.

Table 26 — TermSource attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
status	The status of the term as it is used in this document, relative to its definition in the original document.	M	1	TermOriginStatus
origin	The original document and location where the term definition has been obtained from.	M	1	Citation
modification	Any changes that the definition of the term has undergone relative to the original document, in order to be applicable in this <i>StandardDocument</i> .	O	1	ParagraphBlock

9.12 TermSourceStatus

Table 27 — TermSourceStatus values

Name	Definition
unstructured	The status of a term as it is used in this document, relative to its definition in the original document.
identical	The term definition in this document is identical to that given in the original document.
modified	The term definition in this document has been modified from that given in the original document.

9.13 Nested sections

Both definition and term sections are leaf nodes, and cannot contain subclauses of their own.

Nested definition and term sections should be modelled as `ClauseSections` with subclauses, terminating in definition and term sections.

10 Inline elements

The inline elements of *StandardDocument* adhere to the inline element models from *BasicDocument*.

Specializations and profiles of the *StandardDocument* model may specialize the modelling of inline elements.

10.1 ReferenceToTermbase

Cross-reference to a term defined within a termbase.

Table 28 — ReferenceToTermbase attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
base	Identifier of the termbase.	M	1	String
target	Identifier of the term within the termbase.	M	1	String
text	Text to display for the cross-reference to the term.	M	1	TextElement

10.2 StandocReferenceToIdElement

Cross-reference to an identified element within a *StandardDocument*.

Table 29 — StandocReferenceToIdElement attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
case	Casing of the autogenerated label of the cross-reference (typically the name of the class of element referenced).	0	1	XrefCaseType
droploc	Do not include the autogenerated label of the cross-reference in rendering.	0	1	Boolean
to	If the cross-reference is to a range of text defined as a span between two identified elements, provide the identifier of the end element of the span.	0	1	IdElement

===

10.3 StandardUnorderedList

Unordered list for standards documents.

Table 30 — StandardUnorderedList attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
listitem	List item for standards documents.	0	N	UnorderedCheckableListItem

10.4 UnorderedCheckableListItem

List item for unordered lists for standards documents.

Table 31 — UnorderedCheckableListItem attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
checkbox	Include a checkbox for the list item.	0	1	Boolean
checkedcheckbox	Check the checkbox for the list item.	0	1	Boolean

10.5 StandardDefinitionList

Table 32 — StandardDefinitionList attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
keep-lines-together	Do not permit a page break between the lines of the block in paged media.	0	1	Boolean
keep-with-next	Keep this block on the same page as the following block in paged media.	0	1	Boolean
key	This definition list is the key of a figure or formula.	0	1	Boolean

11 Block

The block elements of *StandardDocument* adhere to the block element *BasicDocument* model and the *StandardDocument* model.

Specializations and profiles of the *StandardDocument* model may specialize the modelling of blocks.

11.1 StandardNoteBlock

Table 33 — StandardNoteBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
keep-lines-together	Do not permit a page break between the lines of the block in paged media.	0	1	Boolean
keep-with-next	Keep this block on the same page as the following block in paged media.	0	1	Boolean
number	Override the numbering of this block in numbering.	0	1	String
subsequence	Define a subsequence for numbering of this block; e.g. if this block would be numbered as 7, but it has a subsequence value of XYZ, this block, and all consecutive blocks of the same class and with	0	1	String

Table 33 — StandardNoteBlock attributes*(continued)*

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
	the same subsequence value, will be numbered consecutively with the same number and in a subsequence: 7a, 7b, 7c etc.			
unnumbered	Do not number this block in rendering.	0	1	Boolean
type	Semantic classification of note.	0	1	String
ul	TODO: enum 's definition	0	N	UnorderedList
ol	TODO: enum 's definition	0	N	OrderedList
dl	TODO: enum 's definition	0	N	DefinitionList
formula	TODO: enum 's definition	0	N	FormulaBlock
sourcecode	TODO: enum 's definition	0	N	SourcecodeBlock
quotes	TODO: enum 's definition	0	N	QuoteBlock

11.2 StandardTableBlock

Table 34 — StandardTableBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
width	Width of the table block in rendering.	0	1	String
number	Override the numbering of this block in numbering.	0	1	String
keep-lines-together	Do not permit a page break between the lines of the block in paged media.	0	1	Boolean
keep-with-next	Keep this block on the same page as the following block in paged media.	0	1	Boolean
colgroup	The widths of the columns in the table.	0	N	TableCol

11.3 TableCol

Table 35 — TableCol attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
width	The width of an individual table column.	0	1	String

11.4 StandardFigureBlock

Table 36 — StandardFigureBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
number	Override the numbering of this block in numbering.	0	1	String
keep-lines-together	Do not permit a page break between the lines of the block in paged media.	0	1	Boolean
keep-with-next	Keep this block on the same page as the following block in paged media.	0	1	Boolean

11.5 StandardSourcecodeBlock

Table 37 — StandardSourcecodeBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
number	Override the numbering of this block in numbering.	0	1	String
keep-lines-together	Do not permit a page break between the lines of the block in paged media.	0	1	Boolean
keep-with-next	Keep this block on the same page as the following block in paged media.	0	1	Boolean

11.6 StandardFormulaBlock

Table 38 — StandardFormulaBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
number	Override the numbering of this block in numbering.	0	1	String
keep-lines-together	Do not permit a page break between the lines of the block in paged media.	0	1	Boolean
keep-with-next	Keep this block on the same page as the following block in paged media.	0	1	Boolean

11.7 StandardExampleBlock

Table 39 — StandardExampleBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
number	Override the numbering of this block in numbering.	0	1	String
keep-lines-together	Do not permit a page break between the lines of the block in paged media.	0	1	Boolean
keep-with-next	Keep this block on the same page as the following block in paged media.	0	1	Boolean

11.8 StandardParagraphBlock

Table 40 — StandardParagraphBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
keep-lines-together	Do not permit a page break between the lines of the block in paged media.	0	1	Boolean
keep-with-next	Keep this block on the same page as the following block in paged media.	0	1	Boolean

11.9 StandardQuoteBlock

Table 41 — StandardQuoteBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
keep-lines-together	Do not permit a page break between the lines of the block in paged media.	0	1	Boolean
keep-with-next	Keep this block on the same page as the following block in paged media.	0	1	Boolean

11.10 StandardAdmonitionBlock

Table 42 — StandardAdmonitionBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
keep-lines-together	Do not permit a page break between the lines of the block in paged media.	0	1	Boolean

Table 42 — StandardAdmonitionBlock attributes*(continued)*

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
keep-with-next	Keep this block on the same page as the following block in paged media.	0	1	Boolean

11.11 AmendBlock

Block expressing a machine-readable change in a document.

Table 43 — AmendBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
change	The type of change described in this block.	M	1	ChangeType
bibLocality	The location in the original document which has undergone the change described in this block. Typically this is given as a clause.	0	N	BibItemLocality
path	The location within the bibLocality where the change applies to, if bibLocality defines a container larger than the scope of the change.	0	1	String
path_end	The end of the span within the bibLocality where the change applies to; applicable to modify or delete.	0	1	String
name	Optional caption of this block.	0	1	String
description	Description of the change described in this block.	0	N	BasicBlock
autonumber	Specification of how blocks of a given class should be autonumbered within an AmendBlock newcontent element.	0	N	AutoNumber
newcontent	New content to be added to the document; applicable to add and modify.	0	N	BasicBlock

11.12 AutoNumber

Specification of how blocks of a given class should be autonumbered within an AmendBlock newcontent element.

Table 44 — AutoNumber attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	The class of block to apply autonumbering to within an AmendBlock newcontent element.	M	1	ElementName
value	The starting value of numbering for the blocks with that class.	M	1	String

11.13 ImageMapBlock

Wrapper around an image file, to specify an image map, with areas of an image being hyperlinked.

Table 45 — ImageMapBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
source	The location of the image file to be updated.	M	1	String
alt	Alternate text to be provided for the image file.	0	1	String
area	Areas of the image to be hyperlinked.	0	N	ImageMapAreaType

11.14 ChangeType

Table 46 — ChangeType values

Name	Definition
unstructured	The type of change described in AmendBlock.
add	
modify	
delete	

11.15 ElementName

Table 47 — ElementName values

Name	Definition
unstructured	The classes of block subject to autonumbering within an AmendBlock newcontent element.
requirement	
recommendation	
permission	
table	
figure	
admonition	
formula	

Table 47 — ElementName values

(continued)

Name	Definition
sourcecode	
example	
note	

11.16 SvgMapBlock

Wrapper around an SVG file, to update its hyperlinks with potentially document-specific links, so that the SVG file can hyperlink to anchors within the document.

Table 48 — SvgMapBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
source	The location of the SVG file to be updated.	M	1	String
alt	Alternate text to be provided for the SVG file.	O	1	String
target	Specification of the cross-references to update the SVG file with.	O	N	SvgTargetType

11.17 SvgTargetType

Specification of (potentially document-specific) cross-references, to overwrite the links within an SVG file, so that the SVG file can hyperlink to anchors within the document.

Table 49 — SvgTargetType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
href	The value of the href attribute in the SVG file to be overwritten.	M	1	String
ReferenceElement	The cross-reference or hyperlink, whose value is to overwrite the value of the href attribute in the SVG file.	M	1	''

11.18 ImageMapAreaType

Specification of an area of an image to be hyperlinked, as part of an image map.

Table 50 — ImageMapAreaType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	Type of an image map area of an image that is hyperlinked.	M	1	ImageMapAreaTypeType

Table 50 — ImageMapAreaType attributes*(continued)*

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
element	Element that this image map area links to.	M	1	ReferenceElement
coords	Polygon defined shape of the image map area.	M	N	ImageMapCoords
radius	Radius-defined shape of the image map area.	O	1	ImageMapRadius

11.19 ImageMapAreaTypeType**Table 51 — ImageMapAreaTypeType values**

Name	Definition
unstructured	The type of area in an image map to be hyperlinked.
rect	Rectangle.
circle	Circle.
ellipse	Ellipse.
poly	Polygon.

11.20 ImageMapCoords

Coordinate point to be used within an image map.

Table 52 — ImageMapCoords attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
x	X-coordinate of the coordinate point.	M	1	Float
y	Y-coordinate of the coordinate point.	M	1	Float

11.21 ImageMapRadius

Radius of a circle area within an image map.

Table 53 — ImageMapRadius attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
x	X-coordinate of the center of a circle area within the image map.	M	1	Float
y	Y-coordinate of the center of a circle area within the image map.	O	1	Float

12 Data types

Data types in *StandardDocument* models are fully inherited from those of the *BasicDocument* model.

Specializations and profiles of the *StandardDocument* model may specialize and extend the modelling of data types.

Annex A (informative)

Profile: ISO standardization deliverables

A.1 General

This profile provides the specialization of the *StandardDocument* document metamodel for ISO standardization deliverables described in ISO/IEC DIR 2 ISO.

The *IsoStandardDocument* model fully covers all ISO standardization deliverables, including International Standards, Guides, Technical Reports, Technical Specifications, International Workshop Agreements, Amendments and Corrigenda.

A.2 Document and section specialization

The *IsoStandardDocument* model extends the *StandardDocument* modelling of the document by requiring the following specific types of section:

- A *Preface*, consisting of a mandatory *Foreword* (a *BasicSection*: CC 36010), and an optional *Introduction* (a *ClauseSection*: [Clause 7](#), since an ISO Introduction can optionally have numbered subsections).
- One or more Clause Sections ([Clause 7](#)).
- Zero or more Annexes ([Clause 7](#)).

Annexes in the *IsoStandardDocument* can optionally contain one or more *Appendixes*, which can be thought of as annexes of the annex section.

NOTE Appendixes are not defined in ISO/IEC DIR 2 ISO, but they appear in the companion standard ISO/IEC DIR 1. For example, Annex SL of ISO/IEC DIR 1 includes Appendix 1, Appendix 2, and Appendix 3; Appendix 2 “High level structure, identical core text, common terms and core definitions” itself contains an Introduction and 10 clauses, including a Terms clause.

The *IsoStandardDocument* model document also includes zero or more *termSources* elements, giving the citations for the documents that the Terms Section draws on for its definitions and repertoire of terms.

The *IsoStandardDocument* model document uses ISO-specific extensions to the CC 6900 metadata about a document ([A.14](#)).

A.3 IsoStandardDocument

ISO Standards document.

Table A.1 — IsoStandardDocument attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
version	Version number of the schema used for this standards document.	M	1	String

Table A.1 — IsoStandardDocument attributes

(continued)

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	Type of standards document representation given in this class instance. Set to “standard” for all standards documents.	M	1	StandardDocumentType
bibdata	Bibliographic description of the document itself, expressed in the Relaton model.	M	1	IsoBibliographicItem
termSources	Citations for sources of term definitions.	0	N	Citation
boilerplate	An optional <i>boilerplate</i> section, intended to appear at the front of the document. It consists of content addressing <i>copyright</i> , <i>license</i> , <i>legal</i> , and <i>feedback</i> concerns.	0	1	BoilerplateType
preface	Zero or more optional <i>preface</i> sections.	M	1	IsoPreface
sections	One or more <i>sections</i> .	M	1	IsoSections
annex	Zero or more <i>annexes</i> .	0	N	AnnexSection
bibliography	Zero to two <i>bibliographies</i> .	0	1	StandardReferencesSection
indexsect	Index of a standards document.	0	1	BasicSection

A.4 IsoAmendmentDocument

ISO Amendment or Technical Corrigenda document.

Table A.2 — IsoAmendmentDocument attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
version	Version number of the schema used for this document.	M	1	String
type	Type of standards document representation given in this class instance. Set to “standard” for all standards documents.	M	1	StandardDocumentType

Table A.2 — IsoAmendmentDocument attributes

(continued)

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
bibdata	Bibliographic description of the document itself, expressed in the Relaton model.	M	1	IsoBibliographicItem
boilerplate	An optional <i>boilerplate</i> section, intended to appear at the front of the document. It consists of content addressing <i>copyright</i> , <i>license</i> , <i>legal</i> , and <i>feedback</i> concerns.	0	1	BoilerplateType
preface	Zero or more optional <i>preface</i> sections.	M	1	IsoPreface
sections	One or more <i>sections</i> . There are no annexes or bibliographies modelled for ISO Amendment or Technical Corrigenda documents.	M	N	IsoAmendmentClause

A.5 IsoSections

Main body of ISO/IEC document.

Table A.3 — IsoSections attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
note	Notes applicable to the entire document (e.g. footnotes on title).	0	N	NoteBlock
admonition	Admonitions applicable to the entire document (appearing before the initial clause).	0	N	AdmonitionBlock
terms	Terms & Definitions clause.	0	1	TermsSection
definitions	Symbols & Abbreviated Terms clause.	0	1	DefinitionsSection
clause	Normal Clauses.	M	N	ClauseSection

A.6 IsoPreface

Prefatory clauses appearing in an ISO/IEC document.

Table A.4 — IsoPreface attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
abstract	Abstract.	0	1	BasicSection
foreword	Foreword.	M	1	BasicSection
introduction	Introduction.	0	1	ClauseSection

A.7 IsoAnnexSection

Annex appearing in ISO/IEC document.

Table A.5 — IsoAnnexSection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
appendix	Appendixes to annex.	0	N	ClauseSection

A.8 IsoAmendmentClause

Clause appearing in an ISO/IEC Amendment or Technical Corrigendum document.

Table A.6 — IsoAmendmentClause attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
amend	Block expressing a machine-readable change in a document.	M	1	AmendBlock

A.9 Block specialization

The *IsoStandardDocument* model extends the *StandardDocument* modelling of blocks for one block definition.

A.10 IsoAdmonitionBlock

Table A.7 — IsoAdmonitionBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	Types of admonition specific to ISO/IEC.	M	1	IsoAdmonitionType

A.11 IsoAdmonitionType

Table A.8 — IsoAdmonitionType values

Name	Definition
unstructured	Types of admonition specific to ISO/IEC.

Table A.8 — IsoAdmonitionType values

(continued)

Name	Definition
danger	Danger.
caution	Caution.
warning	Warning.
important	Important.
safety_precautions	Safety Precautions.

A.12 IsoTermCollection

Term collection specific to ISO/IEC documents.

Table A.9 — IsoTermCollection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
terms	Term subclauses specific to ISO/IEC documents.	0	N	IsoTerm

A.13 IsoTerm

Term subclause specific to ISO/IEC documents.

Table A.10 — IsoTerm attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
preferred	ISO/IEC documents have only a single preferred term name per term subclause.	M	1	LocalizedString
related	ISO/IEC documents do not use related terms.	0	1	RelatedTerm

A.14 Document metadata specialization

The `IsoBibliographicItem` model customizes parts of `BibliographicItem` to fully cover representations of ISO standardization deliverables' metadata.

While `BibliographicItem` is intended as a highly generic model, however, it is useful for the metadata describing the current document to be enriched to provide metadata relevant to the standard class.

`IsoBibliographicItem` is enriched compared to `BibliographicItem` as follows:

- The document identifier is refined to contain three optional parts:
 - a project number,
 - a part number, and
 - a technical committee document number.

- The document title is broken down into:
 - an optional Introductory component,
 - a mandatory Main component, and
 - an optional Part component.
- The document type is constrained to be one of the following:
 - International Standard,
 - Technical Specification,
 - Technical Report,
 - Publicly Available Specification,
 - International Workshop Agreement, and
 - Guide.
- The document status is broken down into:
 - an ISO Document Stage code (ISO International harmonized stage codes),
 - an ISO Document Substage code (ISO International harmonized stage codes), and
 - an Iteration number.
- The editorial group responsible for the document is defined with four components:
 - a mandatory Technical Committee,
 - an optional Subcommittee,
 - an optional Workgroup, and
 - an optional Secretariat.

The Technical Committee, Subcommittee and Workgroup, in turn, are instances of the ISO Subgroup class, defined as having a name, an optional type, and an optional number.

- The document has zero or more source URIs.
- The document has an optional abstract.
- The document has an optional “All Parts” boolean attribute, indicating whether the bibliographic description applies across all Part documents under the same project number (which makes it a multi-document description).

A.15 IsoBibliographicItem

The bibliographical description of an ISO/IEC document.

Table A.11 — IsoBibliographicItem attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
docIdentifier	Identifier of the document.	M	1	IsoDocumentId

Table A.11 — IsoBibliographicItem attributes

(continued)

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
title	Title of the document.	M	1	IsoLocalizedTitle
type	Type of the document.	M	1	IsoDocumentType
status	Publication status of the document.	M	1	IsoDocumentStatus
source	URIs associated with the document.	O	N	TypeUri
abstract	Abstract of the document.	O	N	FormattedString
ext	Extension point of the bibliographical description.	M	1	IsoBibDataExtensionType

A.16 IsoDocumentId

Representation of the identifier for an ISO/IEC document, giving its individual semantic components.

Table A.12 — IsoDocumentId attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
tcDocumentNumber	N-numbers assigned to the document by technical committees.	O	N	Integer
projectNumber	Numerical component uniquely identifying the document.	O	1	Integer
partNumber	Representation in the identifier of the document part, if this is a document part.	O	1	Integer
subpartNumber	Representation in the identifier of the document subpart, if this is a document subpart. Only applicable to IEC documents.	O	1	Integer
amendmentNumber	Representation in the identifier of the type of document amendment, if this is a document amendment.	O	1	Integer
corrigendumNumber	Representation in the identifier of the type of document technical corrigendum, if this is a document technical corrigendum.	O	1	Integer

Table A.12 — IsoDocumentId attributes

(continued)

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
originalyear	Year of publication of the document in its current version or edition.	0	1	Integer

A.17 IsoLocalizedTitle

Title of ISO/IEC document.

Table A.13 — IsoLocalizedTitle attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
titleIntro	Introductory component of title.	0	1	FormattedString
titleMain	Main component of title.	M	1	FormattedString
titlePart	Title of document part, if applicable.	0	1	FormattedString

A.18 IsoDocumentStatus

Publication status of an ISO/IEC document.

Table A.14 — IsoDocumentStatus attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
stage	Publication stage of the ISO/IEC document, expressed as an International Harmonized Stage Code.	M	1	IsoDocumentStageCodes
stage_abbreviation	Standard abbreviation of the publication stage.	0	1	String
substage	Substage code for the ISO/IEC document.	0	1	IsoDocumentSubstageCodes
substage_abbreviation	Standard abbreviation of the publication substage.	0	1	String
iteration	Number of the iteration of the publication stage that the document is in.	0	1	Integer

A.19 IsoProjectGroup

Editorial groups associated with the production of an ISO/IEC document.

Table A.15 — IsoProjectGroup attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
technicalCommittee	Technical committees.	M	N	IsoSubGroup
subcommittee	Subcommittees.	O	N	IsoSubGroup
workgroup	Workgroups.	O	N	IsoSubGroup
secretariat	Secretariat.	O	1	String

A.20 IsoSubGroup

Editorial group set up in ISO or IEC to work on producing documents.

Table A.16 — IsoSubGroup attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	Type of editorial group.	O	1	String
number	Number used to identify editorial group.	O	1	Integer
name	Name of editorial group.	M	1	String

A.21 IsoDocumentType

Table A.17 — IsoDocumentType values

Name	Definition
unstructured	Type of ISO/IEC document.
internationalStandard	International Standard.
technicalSpecification	Technical Specification.
technicalReport	Technical Report.
publiclyAvailableSpecification	Publicly Available Specification.
internationalWorkshopAgreement	International Workshop Agreement.
guide	Guide.
interpretationSheet	Interpretation Sheet.
amendment	Amendment.
technical-corrigendum	Technical Corrigendum.

A.22 IecDocumentCategory

Table A.18 — IecDocumentCategory values

Name	Definition
unstructured	Categories of IEC standards document.

Table A.18 — IecDocumentCategory values

(continued)

Name	Definition
emc	Electromagnetic Compatibility (EMC).
safety	Safety.
environment	Environment.
quality-assurance	Quality Assurance.

A.23 IsoDocumentStageCodes

Table A.19 — IsoDocumentStageCodes values

Name	Definition
unstructured	Stage-level code in the International Harmonized Stage Codes used by ISO/IEC for their documents — see https://www.iso.org/stage-codes.html
00	Preliminary.
10	Proposal.
20	Preparatory.
30	Committee.
40	Enquiry.
50	Approval.
60	Publication.
90	Review.
95	Withdrawal.

A.24 IsoDocumentSubstageCodes

Table A.20 — IsoDocumentSubstageCodes values

Name	Definition
unstructured	Substage-level code in the International Harmonized Stage Codes used by ISO/IEC for their documents — see https://www.iso.org/stage-codes.html
00	
20	
60	
90	
92	
93	
98	
99	

A.25 IsoBibDataExtensionType

Extension point for bibliographical definitions of ISO/IEC documents.

Table A.21 — IsoBibDataExtensionType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
horizontal	The document is a horizontal standard.	0	1	Boolean
editorialgroup	Editorial groups associated with the production of the document.	M	1	IsoProjectGroup
stagename	Name of the publication stage of the document.	0	1	String
category	Category of IEC Standard applicable.	0	1	IecDocumentCategory
updates_document_type	Type of document being updated by this document, if it is an amendment or technical corrigendum.	0	1	DocumentType

Annex B (informative)

Profile: CalConnect standardization deliverables

B.1 General

This profile provides the specialization of the *StandardDocument* document metamodel for CalConnect standardization deliverables described in csd CC/DIR 10002:2019.

The *CsdStandardDocument* model fully covers all CalConnect standardization deliverables, including Standards, Reports, Specifications, Directives, Guides, Amendments and Corrigenda.

B.2 Document and section specialization

The *CsdStandardDocument* model extends the *StandardDocument* modelling of the document by requiring the following specific types of section:

- An optional *Foreword* (a *StandardsHierarchicalSection*, see [Clause 7](#)).
- An optional *Introduction* (a *StandardsHierarchicalSection*, see [Clause 7](#)).
- One or more clause sections (see [Clause 7](#)).
- Zero or more Annexes (see [Clause 7](#)).

B.3 Block specialization

The *Block* model is specialized:

- The definition of *Figure* is altered to allow ASCII artwork as an option for its content, as a string (*pre*). As with the *sourcecode* element (CC 36010), the string should be treated as pre-formatted text, with whitespace treated as significant.

B.4 Inline elements specialization

The *CsdStandardDocument* model extends the *StandardDocument* modelling of inline elements by one element:

- The *keyword* element is added as a subclass of *Text Element* (CC 36010). The element is used to represent code keywords, as with *code* in HTML, and should be rendered in monospace.

B.5 Document metadata specialization

The bibliographic modelling of CalConnect documents is further enriched compared to *BibliographicItem* as follows:

- The document type is constrained to be one of the following:
 - *Standards*,
 - *Reports*,
 - *Specifications*,
 - *Directives*,
 - *Guides*,
 - *Amendments*,
 - *Corrigenda and Administrative*.
- The document status is constrained to be one of the following:
 - *Proposal*,
 - *Working Draft*,
 - *Committee Draft*,
 - *Final Draft*,
 - *Published*,
 - *Withdrawn*.
- The editorial group responsible for the document is defined with a single components: a mandatory Technical Committee. The Technical Committee has a type of either *Technical* or *Provisional*.

B.6 CcBibliographicItem

Bibliographic description of CalConnect document.

Table B.1 — CcBibliographicItem attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
committee	Editorial committee responsible for producing the document.	M	N	CcTechnicalCommittee
type	Type of document.	M	1	CcDocumentType
title	Title of document.	M	1	FormattedString

B.7 CcTechnicalCommittee

CalConnect technical committee.

Table B.2 — CcTechnicalCommittee attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	Type of technical committee.	M	1	CcTechnicalCommitteeType
name	Name of technical committee.	M	1	String

B.8 CcDocumentStatus

Publication status of CalConnect document.

Table B.3 — CcDocumentStatus attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
status	Unitary status description.	M	1	CcStatusType

B.9 CcStatusType

Table B.4 — CcStatusType values

Name	Definition
unstructured	Publication status of CalConnect document.
proposal	Proposal.
working_draft	Working Draft.
committee_draft	Committee Draft.
final_draft	Final Draft.
published	Published.
withdrawn	Withdrawn.

B.10 CcDocumentType

Table B.5 — CcDocumentType values

Name	Definition
unstructured	Types of CalConnect document.
directive	Directive.
standard	Standard.
guide	Guide.
specification	Specification.
amendment	Amendment.
technical_corrigendum	Technical Corrigendum.
report	Report.
administrative	Administrative.

B.11 CcTechnicalCommitteeType

Table B.6 — CcTechnicalCommitteeType values

Name	Definition
unstructured	Type of CalConnect technical committee.
technical	Technical.
provisional	Provisional.

Annex C

(informative)

Concept and concept collection models

C.1 General

This annex provides an informative reference model for management of concepts and concept collections, compliant with ISO 10241-1 and ISO 704.

C.2 Concept

Table C.1 — Concept attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
id	TODO: enum 's definition	0	1	String
designations	TODO: enum 's definition	M	N	Designation
domain	TODO: enum 's definition	0	1	LocalizedString
subject	TODO: enum 's definition	0	1	LocalizedString
usageInfo	TODO: enum 's definition	0	1	LocalizedString
definition	TODO: enum 's definition	0	N	GlossaristDefinition
nonverbalrep	TODO: enum 's definition	0	N	NonVerbRep
notes	TODO: enum 's definition	0	N	GlossaristDefinition
examples	TODO: enum 's definition	0	N	GlossaristDefinition
sources	TODO: enum 's definition	0	N	ConceptSource
related	TODO: enum 's definition	0	N	RelatedConcept

C.3 Designation

C.4 GlossaristDefinition

Table C.2 — GlossaristDefinition attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
content	TODO: enum 's definition	M	N	ParagraphBlock
sources	TODO: enum 's definition	0	N	ConceptSource

C.5 NonVerbRep

Table C.3 — NonVerbRep attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
image	TODO: enum 's definition	0	1	Image
table	TODO: enum 's definition	0	1	Table
formula	TODO: enum 's definition	0	1	FormulaBlock
sources	TODO: enum 's definition	0	N	ConceptSource

C.6 RelatedConcept

Table C.4 — RelatedConcept attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	TODO: enum 's definition	M	1	RelatedConceptType
content	TODO: enum 's definition	M	1	LocalizedString
ref	TODO: enum 's definition	0	1	ReferenceToTermbase

C.7 RelatedConceptType

Table C.5 — RelatedConceptType values

Name	Definition
deprecates	
supersedes	
narrower	

Table C.5 — RelatedConceptType values

(continued)

Name	Definition
broader	
equivalent	
compare	
contrast	
see	

C.8 ConceptSource

Table C.6 — ConceptSource attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
status	TODO: enum 's definition	M	1	ConceptSourceStatus
type	TODO: enum 's definition	M	1	ConceptSourceType
origin	TODO: enum 's definition	M	1	Citation
modification	TODO: enum 's definition	O	1	ParagraphBlock

C.9 ConceptSourceStatus

Table C.7 — ConceptSourceStatus values

Name	Definition
identical	
modified	
restyled	
context-added	
generalisation	
specialisation	
unspecified	

C.10 ConceptSourceType

Table C.8 — ConceptSourceType values

Name	Definition
authoritative	
lineage	

C.11 Designation

Table C.9 — Designation attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
normativeStatus	TODO: enum 's definition	M	1	NormativeStatus
absent	TODO: enum 's definition	0	1	Boolean
geographicArea	TODO: enum 's definition	0	N	Iso3166Code
sources	TODO: enum 's definition	0	N	ConceptSource

C.12 NormativeStatus

Table C.10 — NormativeStatus values

Name	Definition
preferred	
admitted	
deprecated	

C.13 ExpressionDesignation

Table C.11 — ExpressionDesignation attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
text	TODO: enum 's definition	M	1	GlossaristTextElementType
language	TODO: enum 's definition	M	1	Iso639ThreeCharCode
script	TODO: enum 's definition	0	1	Iso15924Code
pronunciation	TODO: enum 's definition	0	N	LocalizedString
grammarInfo	TODO: enum 's definition	0	N	GrammarInfo

C.14 GrammarInfo

Table C.12 — GrammarInfo attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
gender	TODO: enum 's definition	0	N	GrammarGender
isPreposition	TODO: enum 's definition	0	1	Boolean
isParticiple	TODO: enum 's definition	0	1	Boolean
isAdjective	TODO: enum 's definition	0	1	Boolean
isVerb	TODO: enum 's definition	0	1	Boolean
isAdverb	TODO: enum 's definition	0	1	Boolean
isNoun	TODO: enum 's definition	0	1	Boolean

C.15 SymbolDesignation

Table C.13 — SymbolDesignation attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
isInternational	TODO: enum 's definition	0	1	Boolean

C.16 PrefixDesignation

C.17 SuffixDesignation

C.18 AbbreviationDesignation

Table C.14 — AbbreviationDesignation attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
isInternational	TODO: enum 's definition	M	1	Boolean
abbreviationType	TODO: enum 's definition	0	1	AbbreviationType

C.19 AbbreviationType

Table C.15 — AbbreviationType values

Name	Definition
truncation	
acronym	
initialism	

C.20 LetterSymbolDesignation

Table C.16 — LetterSymbolDesignation attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
text	TODO: enum 's definition	M	1	LocalizedString
language	TODO: enum 's definition	M	1	Iso639ThreeCharCode
script	TODO: enum 's definition	O	1	Iso15924Code

C.21 GraphicalSymbolDesignation

Table C.17 — GraphicalSymbolDesignation attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
text	TODO: enum 's definition	O	1	LocalizedString
image	TODO: enum 's definition	O	1	Image
isInternational	TODO: enum 's definition	O	1	Boolean

===

C.22 GrammarGender

Table C.18 — GrammarGender values

Name	Definition
masculine	
feminine	
neuter	
common	

C.23 ManagedConcept

Table C.19 — ManagedConcept attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
related	TODO: enum 's definition	0	N	RelatedConcept
status	TODO: enum 's definition	M	1	ConceptStatus
dates	TODO: enum 's definition	0	N	ConceptDate
localizedConcepts	TODO: enum 's definition	0	1	LocalizedConcept

C.24 ConceptStatus

Table C.20 — ConceptStatus values

Name	Definition
draft	
notValid	
valid	
superseded	
retired	

C.25 LocalizedConcept

Table C.21 — LocalizedConcept attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
language	TODO: enum 's definition	M	1	Iso639ThreeCharCode
script	TODO: enum 's definition	0	1	Iso15924Code
concept	TODO: enum 's definition	M	1	Concept

C.26 ConceptDate

Table C.22 — ConceptDate attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
date	TODO: enum 's definition	M	1	Iso8601Date

Table C.22 — ConceptDate attributes

(continued)

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	TODO: enum 's definition	M	1	ConceptDateType

C.27 ConceptDateType

Table C.23 — ConceptDateType values

Name	Definition
accepted	
amended	
retired	

C.28 RelatedConceptType

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